WORKED SOLUTION

STEP II 2017, Question 2

Pure • Periodic Sequences

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/17-s2-q2

The key idea

Compose the map with itself and simplify: two steps of the recurrence give another map of exactly the same "linear over linear" shape, so four steps are just two steps applied twice. Each time we ask whether $x_{n+k} = x_n$, clearing the denominator produces a quadratic in x_n, and that quadratic factorises. One of its factors is always $x^2 + (b-a)x + 1$, whose roots are precisely the *fixed points* of the recurrence, the values that give a constant sequence. Spotting that is what makes both parts work.

Solution

Write $f(x) = \dfrac{ax-1}{x+b}$, so that $x_{n+1} = f(x_n)$. Then

$$x_{n+2} = \frac{a\dfrac{ax_n-1}{x_n+b} - 1}{\dfrac{ax_n-1}{x_n+b} + b}.$$

Multiplying top and bottom by $x_n + b$ (non-zero, by assumption) gives

$$x_{n+2} = \frac{a(ax_n-1) - (x_n+b)}{(ax_n-1) + b(x_n+b)} = \boxed{\frac{(a^2-1)x_n - (a+b)}{(a+b)x_n + b^2 - 1}}$$

It is worth recording one more fact now. A value x is a fixed point ($f(x) = x$) exactly when $ax - 1 = x^2 + bx$, that is, when

$$q(x) \equiv x^2 + (b-a)x + 1 = 0. \qquad (*)$$

If x_0 is a root of q then $x_1 = x_0$, and so on: the sequence is constant, with period 1.

Part (i)

Suppose the sequence has period 2. Then in particular $x_2 = x_0$. Putting $x = x_0$ in the formula for x_{n+2} and clearing the denominator,

$$\begin{aligned} 0 &= (a^2-1)x - (a+b) - x[(a+b)x + b^2 - 1] \\ &= -(a+b)x^2 + (a^2-b^2)x - (a+b) \\ &= -(a+b)[x^2 + (b-a)x + 1], \end{aligned}$$

using $a^2 - b^2 = (a+b)(a-b)$. So $(a+b)\,q(x_0) = 0$.

> **Ask yourself** Why can't we finish by saying "$q(x_0) = 0$ is only one value of x_0"?

Because x_0 is a single number, and nothing stops it being a root of q. What rules it out is the meaning of $(*)$: if $q(x_0) = 0$ then x_0 is a fixed point, so $x_1 = x_0$ and the sequence has $x_{n+1} = x_n$ for all n. That is period 1, which the definition of period 2 forbids (take $m = 1$). Hence $q(x_0) \neq 0$, and therefore

$$a + b = 0$$

is necessary for period 2.

(Conversely, if $b = -a$ the two-step formula collapses to $x_{n+2} = \dfrac{(a^2-1)x_n}{a^2-1} = x_n$. The case $a^2 = 1$ cannot occur here, since then $x_1 = 1$ when $a = 1$, $b = -1$, or $x_1 = -1$ when $a = -1$, $b = 1$; either way $x_1 + b = 0$, which is excluded.)

Part (ii)

Write $A = a^2 - 1$, $B = b^2 - 1$ and $c = a + b$, so that $x_{n+2} = g(x_n)$ with $g(x) = \dfrac{Ax - c}{cx + B}$. Applying g twice, and again multiplying top and bottom by $cx + B$,

$$x_{n+4} = \frac{A(Ax_n - c) - c(cx_n + B)}{c(Ax_n - c) + B(cx_n + B)} = \frac{(A^2 - c^2)x_n - c(A+B)}{c(A+B)x_n + B^2 - c^2}.$$

Now $x_{n+4} = x_n$ holds exactly when

$$\begin{aligned} 0 &= (A^2-c^2)x_n - c(A+B) - x_n[c(A+B)x_n + B^2 - c^2] \\ &= -c(A+B)x_n^2 + (A^2 - B^2)x_n - c(A+B) \\ &= -(A+B)[c\,x_n^2 + (B-A)x_n + c]. \end{aligned}$$

Since $B - A = b^2 - a^2 = (b-a)(a+b) = (b-a)c$, the bracket is $c\,q(x_n)$, so

$$x_{n+4} = x_n \iff (a^2 + b^2 - 2)\,(a+b)\,q(x_n) = 0.$$

Necessity. Suppose the period is 4. The sequence is not constant, so x_0 is not a fixed point and $q(x_0) \neq 0$. If $a + b = 0$, part (i)'s converse gives $x_2 = x_0$, and indeed $x_{n+2} = x_n$ for

all n, contradicting period 4. So $a + b \neq 0$, and the factorisation with $n = 0$ forces

$$a^2 + b^2 = 2.$$

Sufficiency. Suppose $a^2 + b^2 = 2$. Then $A + B = 0$, so $x_{n+4} = x_n$ for every n. We must check that no smaller m works.

- $a + b \neq 0$: otherwise $a^2 = b^2 = 1$ and, as in part (i), $x_1 + b = 0$, which is excluded.
- There are no real fixed points: the discriminant of q is $(b - a)^2 - 4 = (a^2 + b^2) - 2ab - 4 = -2(1 + ab)$, and $|ab| \leq \frac{1}{2}(a^2 + b^2) = 1$, with $ab = -1$ only when $(a, b) = \pm(1, -1)$, which we have just excluded. So $ab > -1$, the discriminant is negative, and $q(x_n) \neq 0$ for every real x_n. Hence $m = 1$ fails.
- $m = 2$: from part (i), $x_2 = x_0$ would need $(a + b)q(x_0) = 0$, which is false. So $m = 2$ fails.
- $m = 3$: if $x_{n+3} = x_n$ for all n, then $x_{n+1} = x_{(n+1)+3} = x_{n+4} = x_n$ for all n, giving period 1, which we've just ruled out.

Therefore, for real a, b and x_0 (with the denominators non-zero, as the question allows), the sequence has period 4 if and only if

$$\boxed{a^2 + b^2 = 2.}$$

The starting value x_0 plays no part in the condition.

Checks

Take $a = \sqrt{2}$, $b = 0$ and $x_0 = 1$. The sequence runs

$$1, \quad \sqrt{2} - 1, \quad -1, \quad \sqrt{2} + 1, \quad 1, \ \ldots$$

which has period exactly 4. For example, $f(-1) = (-\sqrt{2} - 1)/(-1) = \sqrt{2} + 1$. A computer check confirmed the two-step and four-step formulas exactly for thousands of random rational a, b, x_0. It also confirmed period 4 for about two thousand random points on the circle $a^2 + b^2 = 2$, and found no return after four steps for random (a, b) off that circle.

Common slips

- Concluding from $(a + b)q(x_0) = 0$ that $a + b = 0$ without explaining why $q(x_0) \neq 0$. The fixed-point interpretation is the reason.
- Giving $a^2 + b^2 = 2$ as the answer for period 4 without checking that the period isn't really 1 or 2.
- Losing the factor $a + b$ in the bracket of part (ii), so that the period-2 case $a + b = 0$ wrongly appears to give period 4.

Beyond STEP

A map $x \mapsto \dfrac{\alpha x + \beta}{\gamma x + \delta}$ is a *Möbius transformation*. Composing two such maps corresponds to multiplying their 2×2 matrices, here $\begin{pmatrix} a & -1 \\ 1 & b \end{pmatrix}$. The map has period k exactly when the kth power of the matrix is a multiple of the identity, which is governed by the ratio $(\text{trace})^2/\det = (a+b)^2/(ab+1)$. Period 4 corresponds to the value 2, a "rotation by $\pi/4$" in disguise. Needham's *Visual Complex Analysis* has a beautiful chapter on Möbius transformations, and Strang's *Introduction to Linear Algebra* covers the eigenvalue ideas behind it.

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